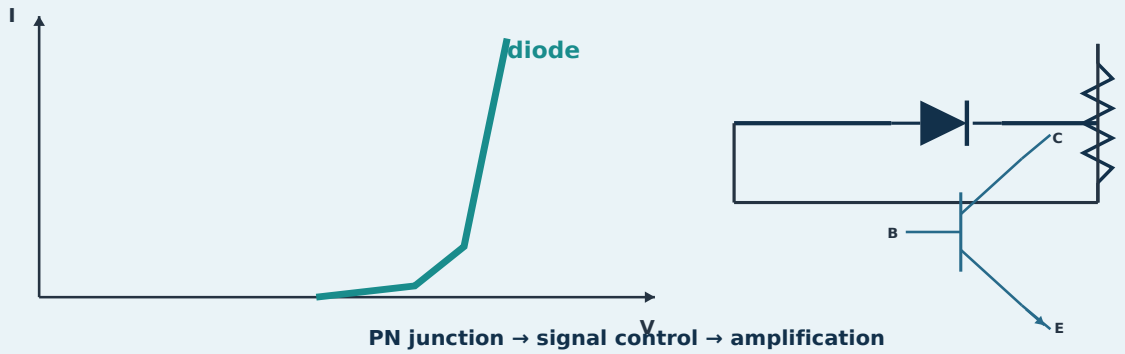
EC 101 • EXAM-READY LEARNING GUIDE

EC 101 — Basics of Electronics & Communication Engineering

Complete, syllabus-aligned learning and examination guide



PN junction → signal control → amplification

A complete, beginner-friendly path through the supplied two-module syllabus: diodes, rectifiers, BJT configurations, characteristics, bias, load lines, CE response, and a clearly separated FET terminology add-on.

MODULE 1	MODULE 2	EXAM TOOLS
Diodes & applications	BJT & amplifier basics	Graphs • numericals • mock

Made from the syllabus and exam-emphasis list supplied in your request. Use it for focused recall, not word-for-word cramming.

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Quick revision map

MODULE 1 · 50% of your revision time PN junction → diode equation and curves → load lines → Zener → HWR/FWR → capacitor filters.	**MODULE 2 · 50% of your revision time** BJT action → CB/CE/CC → characteristics → Q point and bias → DC/AC load lines → CE low-frequency response.	**SYLLABUS TAIL · FET** Channel formation, pinch-off, saturation voltage, and I_D–V_DS output curves. Keep FET saturation distinct from BJT saturation.

Six drawings worth practising closed-book

1 · Diode curve + load line Forward knee, reverse leakage/breakdown, intercepts and Q point.	**2 · Rectifier circuits + waveforms** HWR and FWR before/after C; label f, 2f, PIV and conducting diodes.
3 · Zener regulator Reverse shunt Zener, series resistor, load and current division.	**4 · CB / CE / CC characteristics** Input and output families; state what is held constant.
5 · BJT DC + AC load lines Output curves, Q, cutoff, saturation and AC load slope.	**6 · CE response + FET output** Mark f_L and −3 dB; separately label channel pinch-off and V_DS,sat.

Answer pattern: principle → labelled figure → governing equation → explanation → units and plausibility check. The 50:50 split is your revision plan—not a guaranteed mark distribution.

Modules 1 and 2 • Concepts from first principles • Worked numericals • Characteristic graphs • Draw-it-yourself circuits • Question bank with answers • 50:50 revision plan

Read-this-first scope note. This guide is built from the exact syllabus and the "ECE Exam Important Topics" list in your message. No past paper, marking scheme, textbook, teacher handout, or separate attachment was provided, so I do not pretend to know your examiner's exact wording or guarantee a result. The questions are therefore original practice questions at introductory EC 101 level. Your module list ends with "channel formation, pinch-off and saturation voltage, current-voltage output characteristics," which are field-effect-transistor (FET) terms rather than BJT terms. To avoid a syllabus gap or a terminology mix-up, a clearly labelled FET add-on covers that tail separately.

The 50:50 instruction. Treat Module 1 and Module 2 as equal revision priorities. That is a sensible study allocation from your note, not a promise that the paper will contain exactly 50 marks from each module.

How to use this guide tonight

1. **Understand once:** read the "big picture" and concept sections; do not try to memorize every sentence.
2. **Draw from memory:** reproduce the diode I–V curve, rectifier waveforms, BJT output curves, load lines, and CE frequency-response curve. Compare against the figure checklist after each attempt.
3. **Retrieve, do not just reread:** cover the answer, attempt the question, then mark your own response. Write equations with units and label graph axes.
4. **Finish with the 50:50 practice paper and one-page recall sheet.** Correct errors immediately; those are the topics to review before sleep.

A strong exam answer usually does four things: state the principle; draw a labelled circuit or graph when relevant; write the governing equation and define symbols; then explain the result in words. For a numerical: list known values, convert units, choose the equation, substitute, give units, and check whether the answer is physically plausible.

0. Foundations and notation

0.1 The few electrical ideas everything rests on

- **Voltage** is electrical potential difference, measured in volts (V). It is energy per unit charge.
- **Current** is rate of charge flow, measured in amperes (A). Conventional current points from higher to lower potential through an external circuit; electron motion is opposite.
- **Resistance** opposes current: for an ohmic resistor, $V = IR$. A diode and a transistor are not generally ohmic devices; their voltage-current relationships are nonlinear.
- **KCL:** total current entering a node equals total current leaving it. **KVL:** the signed voltage changes around a closed loop sum to zero.
- **DC** is the steady or average component. **AC** is the time-varying component. A coupling capacitor can pass signal variation while blocking steady DC after it charges.
- For a sinusoid with peak V_m: $V_{rms} = V_m/\sqrt{2}$. For an ideal half-wave rectified sine, average VDC = V_m/π; for an ideal full-wave rectified sine, VDC = $2V_m/\pi$. Do not confuse peak, RMS, and average values.

0.2 Symbols and sign conventions used here

Symbol	Meaning	Usual unit
V_D, I_D	diode anode-to-cathode voltage and forward current	V, A
I_S	diode reverse-saturation current parameter	A
V_T	thermal voltage kT/q	V
n	diode ideality factor	dimensionless
R_D, r_d	diode DC (static) and small-signal (dynamic) resistance	Ω
V_m	peak of the relevant rectifier input sine	V
I_B, I_C, I_E	BJT base, collector, emitter currents	A
V_BE, V_CE	base-emitter and collector-emitter voltage for an NPN convention	V
β, α	common-emitter and common-base current gains	dimensionless
Q point	quiescent DC operating point, before signal is applied	V, A

Polarity note: BJT equations and diagrams below use an NPN transistor unless explicitly marked PNP. Many books draw PNP characteristics with reversed voltage/current signs. The physical operating ideas are the same; follow the sign convention printed in your question.

1. Module 1 — Diodes and applications

1.1 Semiconductor primer: why a diode conducts one way

A pure semiconductor has fewer free carriers than a conductor. Adding a controlled impurity is **doping**:

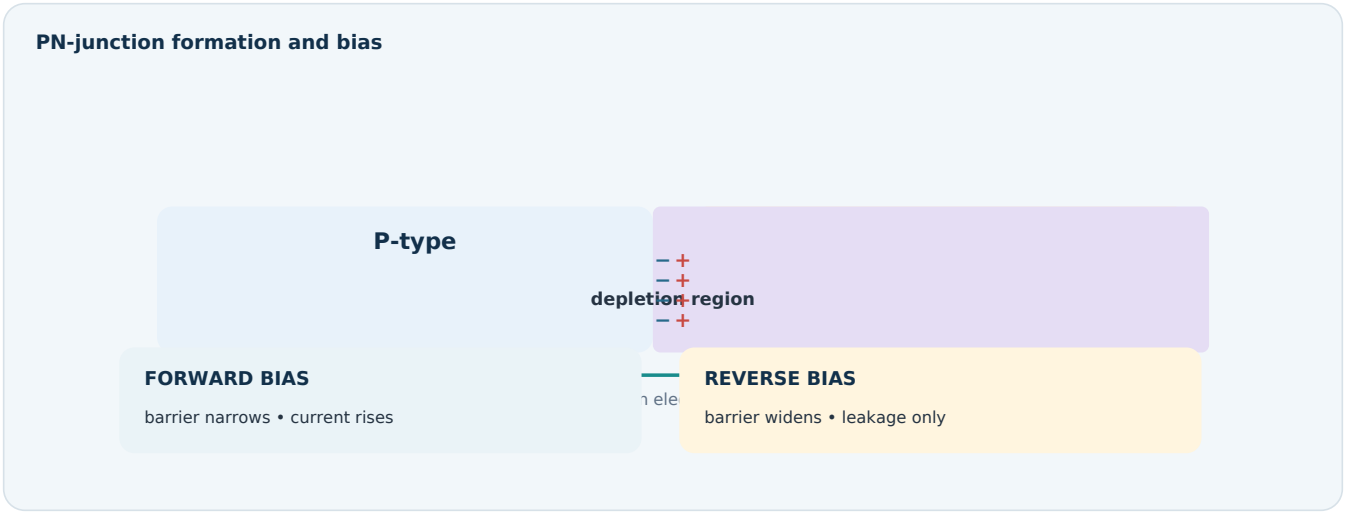
- **N-type:** donor atoms contribute mobile electrons; electrons are majority carriers and holes are minority carriers.
- **P-type:** acceptor atoms create holes; holes are majority carriers and electrons are minority carriers.

A **PN junction** joins P-type and N-type material. Electrons near the junction diffuse from N to P and recombine with holes; holes diffuse the other way. The uncovered ionized dopant atoms form a region with few mobile carriers: the **depletion region**. Its electric field creates a built-in potential barrier. At equilibrium, diffusion and drift balance, so there is no net terminal current.

Forward and reverse bias

- **Forward bias:** connect P/anode to the more positive terminal and N/cathode to the more negative terminal. The barrier and depletion width reduce; majority-carrier injection becomes strong, so current rises rapidly after the knee.
- **Reverse bias:** connect P/anode negative and N/cathode positive. The barrier and depletion width increase. Only a small minority-carrier leakage current flows until breakdown is reached.
- **Conventional diode current** in the forward direction is from anode to cathode. The diode symbol's bar is the **cathode**.

For an ordinary silicon diode, a forward drop around 0.6–0.8 V is a useful circuit approximation at ordinary currents; it is not a fixed universal voltage. Germanium and Schottky devices have different typical drops. For accurate work use the model/data specified in the question.



1.2 Diode current equation and what it means

The Shockley equation models the junction current:

$I_D = I_S [\exp(V_D / (n V_T)) - 1]$, where **$V_T = kT/q$**.

At about 300 K (27 °C), $V_T \approx 25.85$ mV. The diode ideality factor n is often between 1 and 2 in a simple model. I_S is small and strongly device- and temperature-dependent. V_D is positive for forward bias under the convention in this guide.

- When V_D is several nV_T positive, the exponential dominates and I_D rises steeply.
- At $V_D = 0$, the equation gives $I_D = 0$.
- For reverse voltage before breakdown, $\exp(V_D/(nV_T))$ is tiny and $I_D \approx -I_S$.
- The equation's ordinary reverse-bias approximation stops being useful in breakdown.

Numerical method: convert milliamps to amps and millivolts to volts; find V_T at the stated temperature; calculate $x = V_D/(nV_T)$; then evaluate $I_D = I_S(e^x - 1)$. Keep the "−1" if the exponent is not large. A calculator is usually expected for exponentials.

Solved numerical 1 — forward current

A silicon diode has $I_S = 2$ nA and $n = 2$. Find its current at $V_D = 0.60$ V and 300 K.

1. $V_T = 25.85$ mV; $nV_T = 0.05170$ V.
2. $x = 0.60/0.05170 \approx 11.61$; $e^x \approx 1.10 \times 10^5$.
3. $I_D \approx 2 \times 10^{-9}(1.10 \times 10^5 - 1)$ A ≈ **2.20×10^{-4} A = 0.220 mA**.

This is an idealized model answer, not a prediction for every physical diode. The exponential sensitivity explains why a small voltage change can produce a large current change.

Solved numerical 2 — find forward voltage

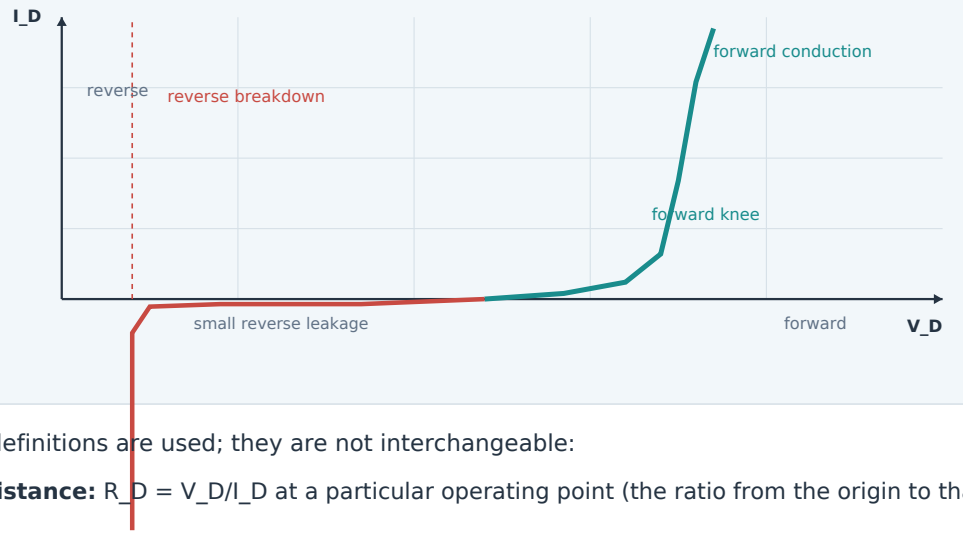
Let $I_S = 10$ nA, $n = 2$, T = 300 K, and $I_D = 1.00$ mA. Rearranging:

$V_D = nV_T \ln(1 + I_D/I_S)$. Here $I_D/I_S = 100{,}000$, so $V_D = 0.05170 \ln(100001) \approx$ **0.596 V**.

1.3 V–I characteristic and diode resistance

The diode's **V–I characteristic** plots diode voltage horizontally and current vertically. Forward current is small below the knee and then rises sharply. Reverse current stays small until the reverse breakdown voltage, where its magnitude rises sharply. Label the axis polarities and identify the breakdown region; an unlabeled curve earns less credit.

Silicon-diode V–I characteristic (illustrative)



Two resistance definitions are used; they are not interchangeable:

- **DC/static resistance:** $R_D = V_D/I_D$ at a particular operating point (the ratio from the origin to that point on the I–V graph).
- **AC/dynamic/small-signal resistance:** $r_d = dV_D/dI_D$ at the operating point (the local slope resistance). From the Shockley model in forward conduction, **$r_d \approx nV_T/I_D$**. At 1 mA, 300 K, $n = 1$, $r_d \approx 25.9\ \Omega$; for $n = 2$, $\approx 51.7\ \Omega$.

Use the dynamic resistance only for a sufficiently small signal around the chosen DC operating point. A large signal sweeps across the nonlinear curve and cannot be represented by one constant r_d.

1.4 Temperature dependence and diode capacitance

Temperature

At fixed forward current, a silicon diode's forward voltage typically falls by roughly 2 mV/°C near ordinary operating conditions; the exact slope depends on current and device. Reverse saturation/leakage current generally rises strongly with temperature (a rough rule of thumb is that it can approximately double for each 10 °C, but this is not universal). Therefore:

- a fixed forward voltage does **not** imply a temperature-independent current;
- a diode or transistor bias network may drift as temperature changes;
- check the device's data sheet/rating in a real design.

Junction capacitance

A diode also stores charge, so it has capacitance:

1. **Transition/depletion/junction capacitance C_T:** important mainly under reverse bias. The depletion region acts like the dielectric between charge regions. Increasing reverse bias widens the depletion region and normally decreases C_T. A common model is **$C_T = C_{J0}/(1 + V_R/\Phi_0)^m$**, with device-dependent parameters.
2. **Diffusion capacitance C_D:** important in forward bias because injected minority carriers are stored near the junction. A small-signal approximation is **$C_D \approx \tau_T I_D/(nV_T)$**, where τ_T is a transit/storage time.

Total junction behavior depends on bias and frequency. When an exam asks "diode capacitance," identify which mechanism and bias region you mean; do not describe the reverse-biased depletion capacitance as the dominant forward-bias storage effect.

1.5 DC and AC load lines; operating point

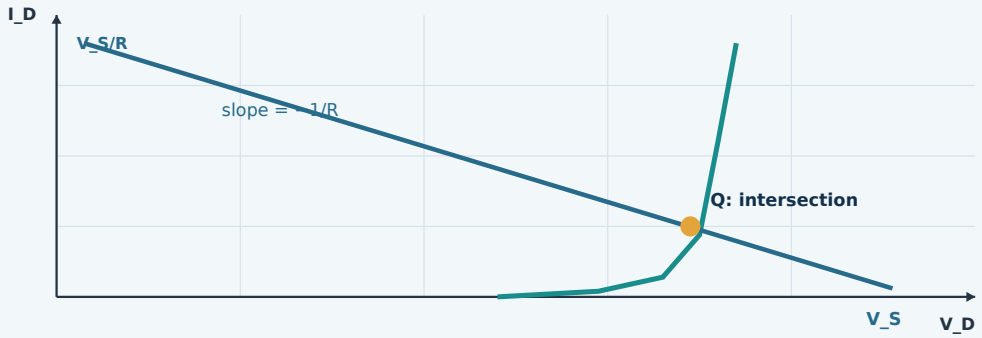
A load line is the straight-line constraint imposed by the external source and resistor. For a source V_S, series resistor R, and diode:

$V_S = I_D R + V_D$, hence **$I_D = (V_S - V_D)/R$**.

Its intercepts are ($V_D = V_S$, $I_D = 0$) and ($V_D = 0$, $I_D = V_S/R$); its slope on an I-versus-V graph is $-1/R$. The intersection with the diode's nonlinear characteristic gives the **DC operating point** (Q point). The graphical method and the circuit equation must agree.

For a small AC change, linearize around Q. The AC load line passes through Q; its slope is set by the effective small-signal load, often $-1/R_{ac}$. It describes signal variation around the bias point, while the DC load line describes the steady source-and-load constraint. In an actual circuit identify which resistors are seen by DC and by AC; coupling capacitors can make those loads different.

Load line and diode operating point



Solved numerical 3 — resistive load line

A 10 V DC source feeds a silicon diode through 1.0 kΩ. Using the constant-drop model $V_D = 0.70$ V, $I_D = (10 - 0.70)/1000 =$ **9.3 mA**. The ideal load-line intercepts are 10 V on the voltage axis and 10 mA on the current axis; the practical diode operating point is near (0.70 V, 9.3 mA). The resistor dissipates $I^2R \approx 86.5$ mW, so a quarter-watt resistor is adequate for this nominal calculation.

1.6 Breakdown: Zener and avalanche

Breakdown is a sharp rise in reverse current when the reverse electric field becomes sufficiently large. It is not automatically destructive: it is safe only when the external circuit limits current and the device stays within its ratings.

Feature	Zener-dominant breakdown	Avalanche-dominant breakdown
Physical mechanism	Quantum-mechanical tunnelling across a very strong field	Impact ionization: accelerated carriers create additional electron–hole pairs
Typical junction	Heavily doped; narrow depletion region	More lightly doped; wider depletion region
Typical voltage trend	More common at lower breakdown voltages	More common at higher breakdown voltages
Temperature coefficient (rule of thumb)	Often negative	Often positive
Practical caveat	Real devices may involve both mechanisms; near about 5–6 V the coefficients can partly cancel	Exact mechanism/voltage depends on construction and rating

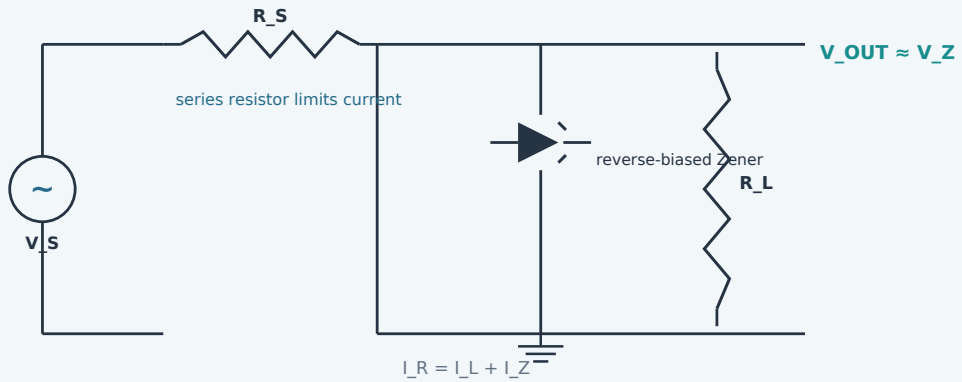
Do not define the difference as merely “low voltage versus high voltage”: the **physical mechanism and junction structure** are the key distinction. The label “Zener diode” is also commonly used for a practical reverse-breakdown voltage-reference diode even if avalanche contributes.

Zener diode as a shunt regulator

Connect the Zener in reverse across the load and add a **series resistor** from the source. The Zener holds the load voltage near V_Z while it remains in its regulation-current range. The resistor drops the excess voltage and limits current. At the design worst case, verify:

- $I_R = (V_S - V_Z)/R_S$;
- $I_R = I_L + I_Z$;
- I_Z must remain above the specified knee/minimum regulation current and below its maximum rating;
- $P_Z = V_Z I_Z$ and $P_R = I_R^2 R_S$ must be within ratings.

Zener shunt regulator



Solved numerical 4 — nominal Zener resistor

A 12 V source feeds a 5.1 V Zener regulator and 1.0 kΩ load. Require nominal $I_Z \geq 5$ mA. Load current $I_L = 5.1/1000 = 5.1$ mA. Thus $R_S \leq (12 - 5.1)/(5.1\text{ mA} + 5\text{ mA}) \approx$ **683 Ω**. Choose 680 Ω (assuming an appropriate power rating and tolerances). Nominal $I_R = 6.9/680 = 10.15$ mA; $I_Z \approx 10.15 - 5.1 =$ **5.05 mA**. $P_Z \approx 25.8$ mW and $P_R \approx 70$ mW nominally. A real design must also check supply extremes, load extremes, resistor tolerance, and the Zener’s maximum rated power.

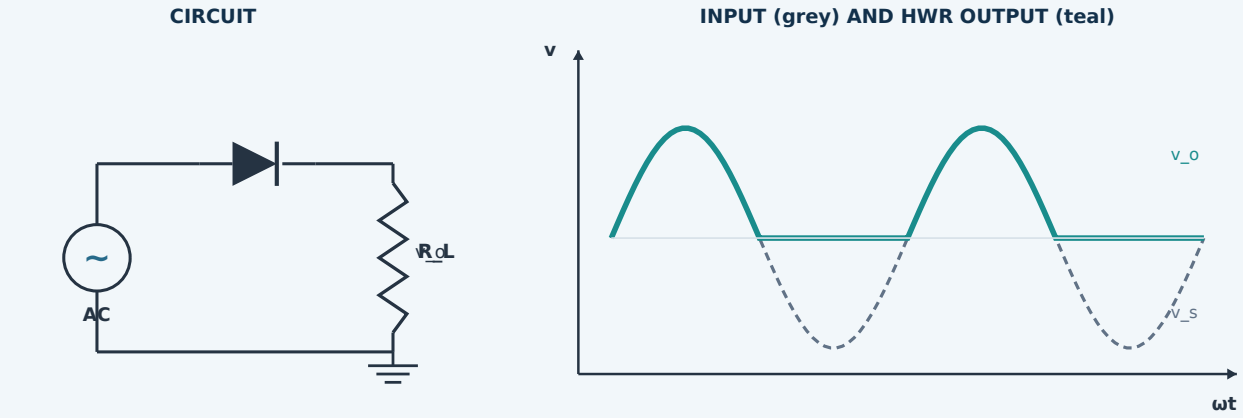
1.7 Diode rectifiers: turn AC into one-polarity output

A rectifier uses the diode's one-way conduction to produce a unidirectional output. Unless stated otherwise, the standard values below assume **ideal diodes** and a sinusoidal input; real diode drops and source resistance reduce the output.

Half-wave rectifier (HWR)

One diode conducts on one input half-cycle and blocks the other. For a resistive load, the ideal output is a positive half-sine followed by zero (or a negative half-sine if the diode is reversed).

Half-wave rectifier: circuit and waveform



For input $v_s = V_m \sin(\omega t)$:

- Average/DC output: **$V_{DC} = V_m/\pi \approx 0.318V_m$**.
- RMS output: **$V_{RMS} = V_m/2$**.
- Ripple frequency: **$f_r = f$** (same as supply frequency).
- Peak inverse voltage (PIV) of the diode: **V_m**.
- Maximum ideal rectification efficiency: about **40.6%**; ripple factor about **1.21**.

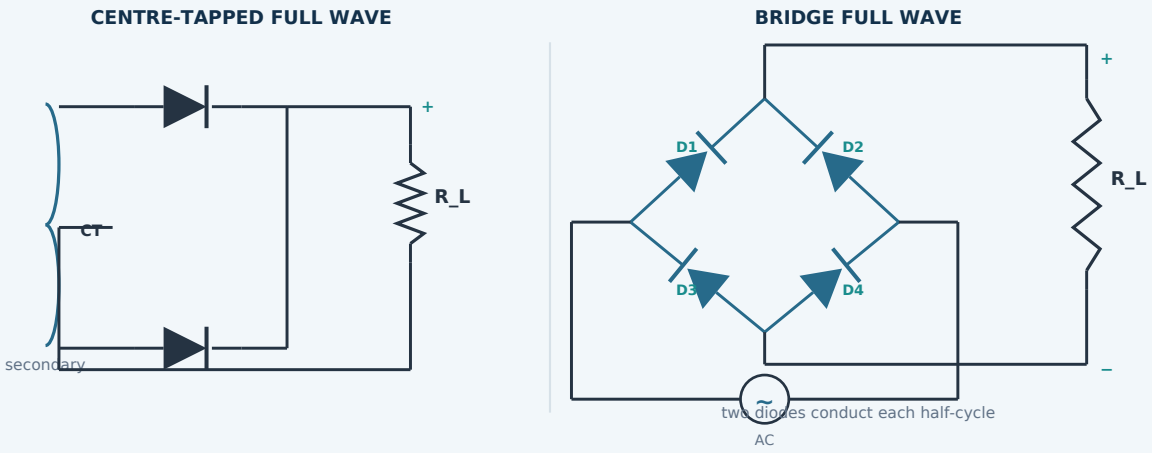
It is simple and uses one diode, but wastes a half-cycle and has larger ripple than a full-wave circuit.

Full-wave rectifier (FWR): centre-tapped and bridge

Both input half-cycles produce the same load-current direction. Two common arrangements:

- **Centre-tapped:** a centre-tapped transformer and two diodes; one diode conducts on each half-cycle. The PIV per diode is $2V_m$ when V_m is the peak voltage of **one half-secondary**.
- **Bridge:** four diodes; two conduct at a time. No centre tap is required. The PIV per diode is V_m when V_m is the peak voltage across the complete bridge input. Two forward diode drops are in the current path.

Full-wave rectifiers: centre-tapped and bridge



For an ideal resistive load (V_m is the rectified sine peak):

- Average/DC output: **$V_{DC} = 2V_m/\pi \approx 0.637V_m$**.
- RMS output: **$V_{RMS} = V_m/\sqrt{2} \approx 0.707V_m$**.
- Ripple frequency: **$f_r = 2f$**.
- Ideal maximum rectification efficiency: about **81.2%**; ripple factor about **0.482**.

A full-wave rectifier uses both half-cycles, so its average output is twice the half-wave average for the same V_m, and its ripple is easier to filter. **PIV comparisons are only fair when V_m is defined for the relevant winding/input.**

Rectifier with a capacitor filter

Place a capacitor in parallel with the load. Near each waveform peak, the rectifier charges it; between peaks the diode turns off and the capacitor supplies the load. The output becomes a high DC level with a sawtooth-like ripple. The same filtering idea applies to HWR and FWR: an HWR capacitor is refreshed once per input cycle ($f_r = f$), while an FWR capacitor is refreshed twice per cycle ($f_r = 2f$). For the same C, load current, and mains frequency, the FWR therefore has about half the ripple estimate of the HWR. A larger C, lighter load (larger R_L), or higher ripple frequency reduces ripple.

HWR and FWR: with and without capacitor filters



For small ripple, a useful approximation is:

$\Delta V_{pp} \approx I_L/(f_r C)$, where ΔV_{pp} is peak-to-peak ripple, I_L is load current, f_r = f for HWR or 2f for FWR, and C is in farads. A rough average is $V_{DC} \approx V_{peak,charged} - \Delta V_{pp}/2$. This assumes the capacitor discharges approximately linearly between charging peaks; it is an approximation, not an exact waveform solution.

For a bridge with a capacitor, $V_{peak,charged}$ is roughly $V_m - 2V_F$; for a centre-tapped full-wave circuit it is roughly the half-winding $V_m - V_F$. Without a capacitor, do **not** use "peak minus ripple" as the average; use the rectified-waveform average formulas above.

Solved numerical 5 — compare rectifier averages

A 12 V RMS sinusoid feeds ideal rectifiers. $V_m = \sqrt{2} \times 12 = 16.97$ V.

- HWR: $V_{DC} = 16.97/\pi =$ **5.40 V**; $V_{RMS} = 16.97/2 =$ **8.49 V**.
- FWR: $V_{DC} = 2(16.97)/\pi =$ **10.80 V**; $V_{RMS} = 16.97/\sqrt{2} =$ **12.00 V**.

These are ideal, **unfiltered** outputs. A filtered output sits nearer the peak and depends on load, C, frequency, winding resistance, and diode drops.

Solved numerical 6 — capacitor ripple estimate

A full-wave rectifier on 50 Hz mains supplies approximately 20 mA through a 1000 µF capacitor. $f_r = 100$ Hz and $C = 0.001$ F, so $\Delta V_{pp} \approx 0.020/(100 \times 0.001) =$ **0.20 V peak-to-peak**. The small-ripple assumption is reasonable only if the capacitor voltage is much larger than 0.20 V.

1.8 Module 1 exam traps and mini-check

- A forward diode drop is **not** a universal 0.7 V; state the model you use.
- Static resistance is V/I; dynamic resistance is local $\Delta V/\Delta I$. Do not swap them.
- In a rectifier, DC average, RMS, and peak are different quantities.
- An HWR ripple frequency is f; an FWR ripple frequency is 2f.
- A bridge has two conducting diode drops; a centre-tapped FWR has one conducting diode drop.
- A Zener needs a series current-limiting resistor. "Zener clamps voltage" alone is not a complete regulator explanation.
- In diode equation calculations, convert nA, mA, mV, and µF correctly.

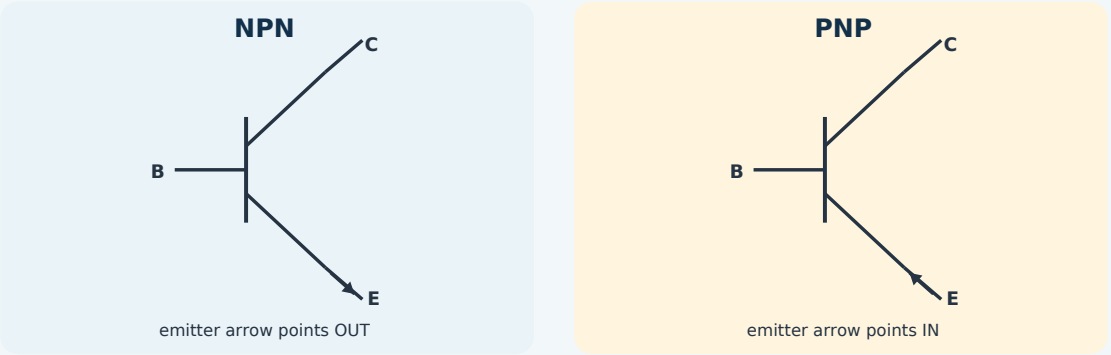
30-second retrieval: Why does forward bias narrow the depletion region? Which capacitance dominates reverse bias? What two intercepts define a diode load line? How does a full-wave capacitor filter differ from an unfiltered waveform? Answer without looking before moving on.

2. Module 2 — Bipolar junction transistors

2.1 Construction and basic transistor action

A BJT has three doped regions and two junctions: emitter–base (EB) and collector–base (CB). The two polarities are **NPN** and **PNP**. The emitter is heavily doped to inject carriers; the base is very thin and lightly doped so most injected carriers cross it; the collector collects carriers and is designed to withstand voltage and dissipate heat.

NPN and PNP transistor symbols



In normal **forward-active** NPN operation, EB is forward biased and CB reverse biased. The emitter injects electrons into the thin base. A small fraction recombines there and is replenished by base current; most electrons are swept into the collector by the reverse-biased CB field. Thus a small base current controls a larger collector current. PNP action is analogous with holes and reversed polarities.

Current relations and gains

For magnitudes in forward-active operation:

- **$I_E = I_C + I_B$** (KCL).
- **$\alpha = I_C/I_E$**, usually slightly less than 1.
- **$\beta = I_C/I_B$**, typically much larger than 1 but device- and operating-point-dependent.
- **$\beta = \alpha/(1 - \alpha)$**; **$\alpha = \beta/(\beta + 1)$**; **$I_C \approx \beta I_B$** in the active region.
- Common-collector current gain **$\gamma = I_E/I_B = \beta + 1$**.

Example: if $\beta = 99$, then $\alpha = 99/100 = 0.99$. A 20 µA base current would ideally control $I_C \approx 1.98$ mA in active operation. It cannot continue doing so once the transistor is driven into saturation.

Four operating regions

Region	EB junction	CB junction	Approximate behavior / use
Cutoff	Not forward biased	Reverse biased in usual NPN switch convention	I_B and I_C are nearly zero; switch OFF
Forward active	Forward biased	Reverse biased	$I_C \approx \beta I_B$; amplification
Saturation	Forward biased	Forward biased	Fully ON switch; I_C is limited by circuit, not simply βI_B; $V_{CE(sat)}$ is small but nonzero
Reverse active	Reverse biased	Forward biased	Collector/emitter roles effectively exchanged; rarely used; poor gain

A transistor is not “in active mode” merely because it has a base current. Check the junction biases and the operating point. Typical silicon V_{BE} near 0.7 V is a first approximation, not an exact law.

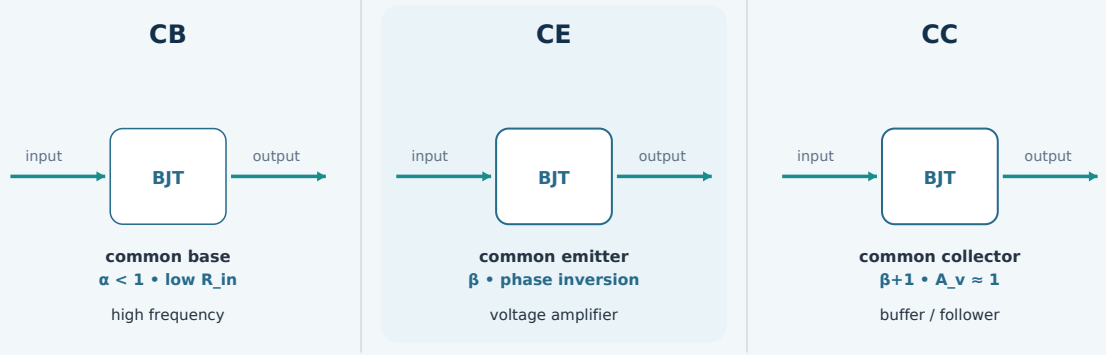
2.2 CB, CE, and CC configurations

A “common” terminal is shared by input and output circuits (often at AC ground). Input and output port definitions matter. The comparison below is the high-yield memorization table.

Property	Common base (CB)	Common emitter (CE)	Common collector (CC / emitter follower)
Common terminal	Base	Emitter	Collector
Input port	Emitter–base	Base–emitter	Base–collector (collector common)
Output port	Collector–base	Collector–emitter	Emitter–collector
Current gain	$\alpha < 1$	β, usually high	$\gamma = \beta + 1$, high
Voltage gain	Often high	Often high	Approximately 1, slightly below 1
Input resistance	Low	Medium (application-dependent)	High
Output resistance	High	High / medium-high	Low
Voltage phase	No inversion	180° inversion	No inversion
Typical use	High-frequency / low-input-resistance stage	General-purpose voltage/power gain	Buffer and impedance matching

All of these are qualitative tendencies; actual values depend on the transistor, bias, and surrounding circuit. “CC voltage gain near one” does not mean “no current gain.”

Three BJT configurations: input, output, common terminal

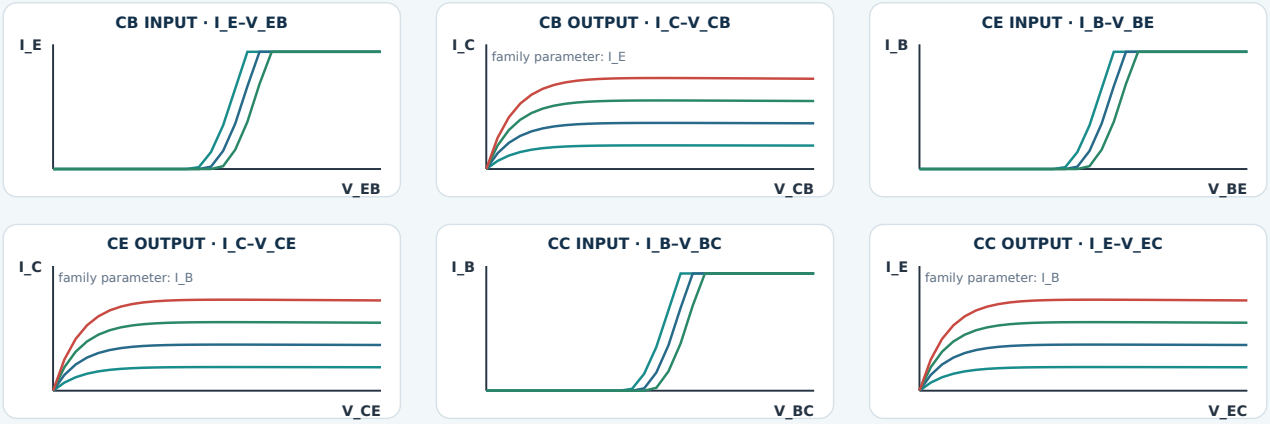


Characteristic curves: what to plot and how to read them

Characteristics are usually plotted as families because one terminal variable is held at several values. Identify which variable is held constant.

- **CB input:** I_E versus V_{EB} (or V_{BE} under a stated sign convention), for fixed V_{CB}. It is diode-like and has low input resistance.
- **CB output:** I_C versus V_{CB}, for fixed I_E. Active-region curves are relatively flat; I_C is slightly less than I_E. The low-voltage knee marks saturation.
- **CE input:** I_B versus V_{BE}, for fixed V_{CE}. It resembles a forward-biased junction; changing V_{CE} can shift the curve slightly.
- **CE output:** I_C versus V_{CE}, for fixed I_B. Curves show cutoff, a saturation knee, and a comparatively flat active region; at very high V_{CE} the transistor can break down.
- **CC input/output:** same transistor physics with collector common. Commonly the input is I_B versus V_{BC} at fixed V_{EC}, and the output is I_E versus V_{EC} for fixed I_B (labels/signs vary by text). Output curves resemble CE curves but plot emitter current. Always copy the axes requested by your course/question.

Typical BJT characteristic families for CB, CE, and CC



Illustrative curves; sign conventions and held-constant labels may vary by text.

2.3 DC biasing and the Q point

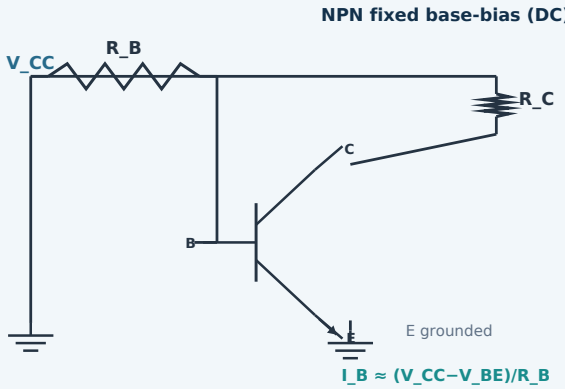
Biasing establishes DC junction voltages and currents so a transistor begins at a useful operating point, typically in forward-active mode for an amplifier. The **Q point** is (I_CQ, V_CEQ) for a CE circuit. If Q is too near cutoff or saturation, a signal clips on one side. A roughly central Q point often provides useful symmetrical swing in a simple supply, but the best point depends on the circuit and load.

Fixed-bias (base-bias) circuit

For an NPN with emitter grounded, R_B from V_CC to base, and R_C from V_CC to collector:

- I_B ≈ (V_CC − V_BE)/R_B.
- I_C ≈ βI_B, provided the result is in active mode.
- V_CE = V_CC − I_C R_C.
- DC load-line endpoints: (V_CE = V_CC, I_C = 0) and approximately (V_CE = 0, I_C = V_CC/R_C).

Fixed-bias NPN circuit



Strength: very simple. **Weakness:** I_C changes significantly with β, V_BE, and temperature; fixed bias is therefore poorly stabilized. Always check that computed V_CE is above saturation and within rating before trusting I_C = βI_B.

Emitter-resistor bias / emitter feedback

Adding R_E introduces negative feedback. If I_C rises, I_E rises, V_E rises, and V_BE = V_B − V_E falls; the base-emitter drive and current are pushed back down. This stabilizes bias against β and temperature changes.

With a base voltage source V_BB and base resistor R_B, a useful equation is:

I_B ≈ (V_BB − V_BE)/[R_B + (β + 1)R_E], then I_C ≈ βI_B and V_E ≈ I_E R_E.

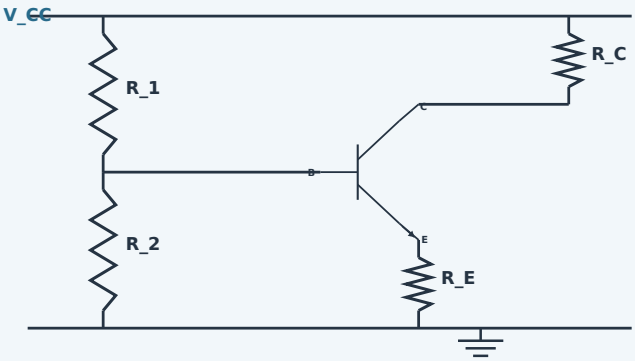
If the base is held stiffly at a known V_B, the simpler estimate is **I_E ≈ (V_B − 0.7)/R_E** and I_C ≈ I_E when β is large. Check units and the active-region assumption.

Voltage-divider “self-bias”

Many introductory texts call the R_1–R_2 base-divider plus R_E network **self-bias**. Other syllabi use the term differently; show the circuit and state the stabilization mechanism so the meaning is clear. Replace the divider by its Thevenin equivalent:

- V_TH = V_CC R_2/(R_1 + R_2), with R_2 from base to ground.
- R_TH = R_1 || R_2.
- I_B ≈ (V_TH − V_BE)/[R_TH + (β + 1)R_E].
- If divider current is much larger than I_B, V_B ≈ V_TH; I_E ≈ (V_B − V_BE)/R_E. This makes the bias less sensitive to β.

Voltage-divider self-bias with emitter feedback



R_E feedback stabilizes I_C

Solved numerical 7 — fixed bias

An NPN has V_CC = 12 V, R_B = 470 kΩ, R_C = 2.2 kΩ, β = 100, V_BE = 0.70 V, and grounded emitter.

I_B = (12 − 0.70)/470 kΩ = **24.0 μA**. Estimated I_C = 100 × 24.0 μA = **2.40 mA**. V_CE = 12 − (2.40 mA)(2.2 kΩ) = **6.72 V**. This is well above a small saturation voltage, so the active-region estimate is self-consistent for this simplified calculation.

Solved numerical 8 — divider bias (Thevenin method)

Let V_CC = 12 V, R_1 = 47 kΩ, R_2 = 10 kΩ, R_C = 2 kΩ, R_E = 1 kΩ, β = 100, and V_BE = 0.70 V. V_TH = 12(10/57) = 2.105 V. R_TH = 47 kΩ || 10 kΩ = 8.25 kΩ. Then I_B = (2.105 − 0.70)/[8.25 kΩ + 101(1 kΩ)] ≈ **12.8 μA**; I_C ≈ **1.28 mA**; I_E ≈ 1.29 mA; V_E ≈ 1.29 V; V_C = 12 − (1.28 mA)(2 kΩ) ≈ 9.44 V; V_CE ≈ **8.15 V**. Rounded values differ slightly depending on whether I_E = (β+1)I_B is retained.

2.4 DC and AC load-line analysis

DC load line

For a CE stage with R_C and R_E, KVL gives approximately:

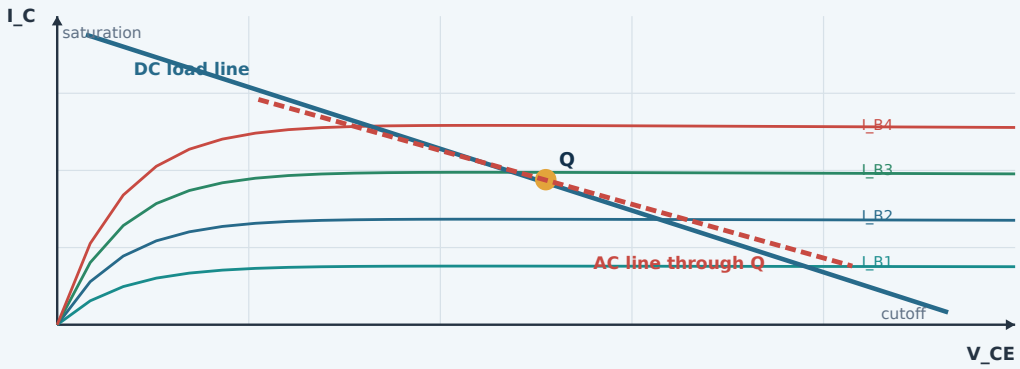
V_CE = V_CC − I_C R_C − I_E R_E.

For large β, I_E ≈ I_C, so V_CE ≈ V_CC − I_C(R_C + R_E). If R_E is bypassed for AC, it remains in the **DC** equation. The intersection of the DC load line and the transistor output characteristic selected by I_B (or bias network) is the Q point.

AC load line

At midband, coupling capacitors are often approximated as shorts and the supply is AC ground. An external load R_L may appear in parallel with R_C, so R_ac ≈ R_C || R_L. The AC load line passes through Q and has slope approximately **−1/R_ac** on an I_C-versus-V_CE graph. Its extent is limited by cutoff on one side and saturation/distortion on the other. It predicts signal excursions; the DC load line is still needed to find the bias point.

BJT output curves with DC and AC load lines



Exam method: (1) draw axes I_C vertical, V_CE horizontal; (2) mark cutoff at I_C ≈ 0; (3) mark saturation near V_CE(sat); (4) draw the DC line from the supply/load endpoints; (5) mark Q; (6) draw the AC line through Q with the AC load slope; (7) label the two lines and explain which circuit conditions each represents.

2.5 Low-frequency response of a common-emitter amplifier

A CE amplifier commonly has an input coupling capacitor, an output coupling capacitor, and an emitter bypass capacitor. These capacitors do not behave as perfect shorts at low frequency. Their reactance is **X_C = 1/(2πfC)**, so lower f means larger reactance:

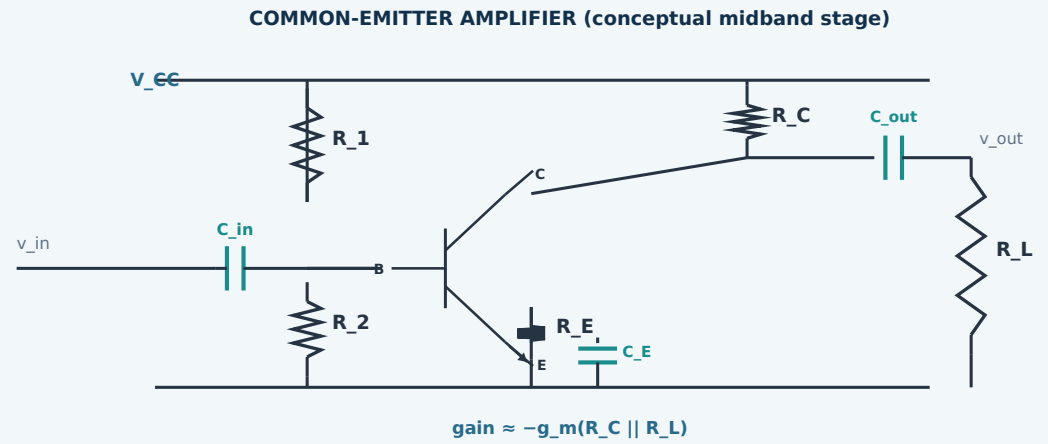
- Input coupling capacitor: attenuates the input signal at low f.
- Output coupling capacitor: attenuates the signal delivered to R_L at low f.
- Emitter bypass capacitor: if its reactance is large, R_E is not fully bypassed for AC; emitter degeneration increases and gain falls.

At **midband**, coupling/bypass capacitors act approximately as shorts and transistor parasitic capacitances are not yet dominant. At **low frequency**, gain rises with frequency toward midband. At high frequency, internal capacitances eventually reduce gain; high-frequency behavior is included for context even though your listed emphasis is low-frequency response.

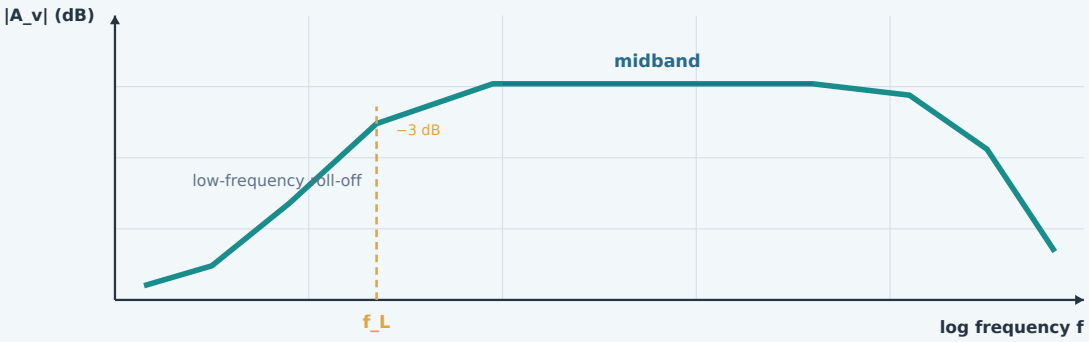
For a bypassed-emitter small-signal CE stage, a useful midband estimate is **A_v ≈ −g_m(R_C || R_L)**, where g_m ≈ I_C/V_T at room temperature. An alternative approximation is A_v ≈ −(R_C || R_L)/r_e. The minus sign means approximately 180° voltage phase inversion. In dB, voltage gain = 20 $\log_{10}$|A_v|.

A coupling capacitor's approximate lower corner is f_c ≈ 1/(2πR_eq C), using the resistance seen by that capacitor with independent sources set to zero. The actual lower cutoff f_L is determined by the combined low-frequency poles; when the corners are well separated, the highest individual corner often dominates. At f_L the gain is about 0.707 of midband voltage gain (−3 dB). Do not automatically use the same R_eq for all three capacitors.

Common-emitter amplifier and its low-frequency capacitors



CE amplifier frequency response (schematic)



2.6 Module 2 exam traps and mini-check

- I_E = I_C + I_B. β = I_C/I_B; α = I_C/I_E. For finite base current, α is slightly below one.
- I_C ≈ βI_B applies in forward-active operation—not after saturation.
- CE has a 180° voltage phase inversion; CB and CC do not.
- CC has high current gain and voltage gain near one; it is a buffer, not a voltage-gain stage.
- DC and AC load lines can differ because coupling capacitors and load resistors affect AC but not the DC bias.
- Emitter resistance stabilizes bias through negative feedback. Fixed bias does not.
- A characteristic graph must say what is held constant for each curve.

30-second retrieval: In active NPN operation, which junction is forward biased? What is the physical reason I_C is controlled by base drive? Why does R_E stabilize bias? Which configuration inverts phase? What does Q mean?

3. Syllabus-tail clarification — FET channel and pinch-off

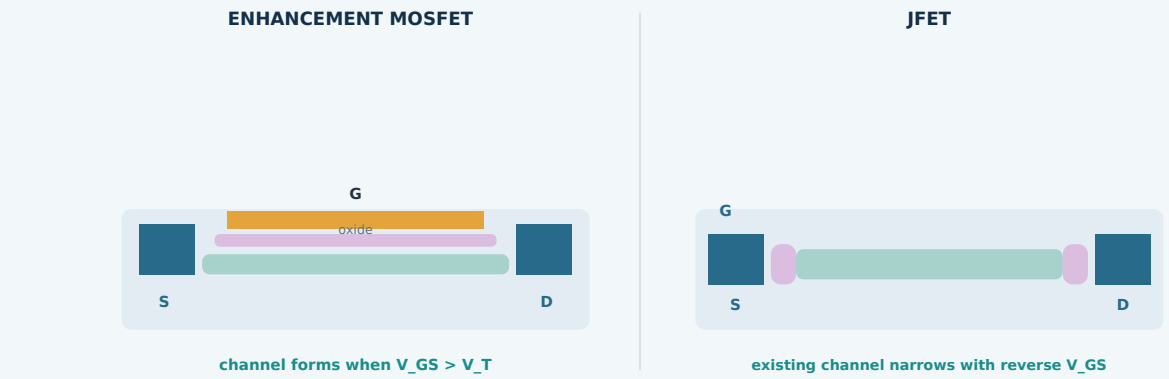
Your pasted list places “Idea of Channel Formation, Pinch-Off and saturation Voltage, Current-Voltage Output Characteristics” after BJT topics. **Those are FET vocabulary**, not normal BJT characteristic terms. Treat this section as the most likely intended brief FET coverage; use your teacher’s exact device type if it has been named elsewhere. Do not explain BJT collector-current saturation as FET channel pinch-off.

3.1 What a field-effect transistor does

A FET uses an electric field at the gate to control a channel between source and drain. The gate draws very little steady current in an insulated-gate MOSFET; a JFET gate is reverse biased in normal operation. For an **n-channel enhancement MOSFET**, at V_GS below threshold V_T there is no strong inversion channel and I_D is very small. Raising V_GS above V_T forms an electron-rich channel; applying V_DS drives drain current. A p-channel device has opposite polarities and carrier type.

A **JFET differs**: it has a conducting channel at V_GS = 0, and reverse gate bias widens depletion regions and narrows the channel. State whether a question asks about enhancement MOSFET or JFET; their channel-formation descriptions are not interchangeable.

Channel formation: enhancement MOSFET versus JFET



3.2 Pinch-off and saturation voltage

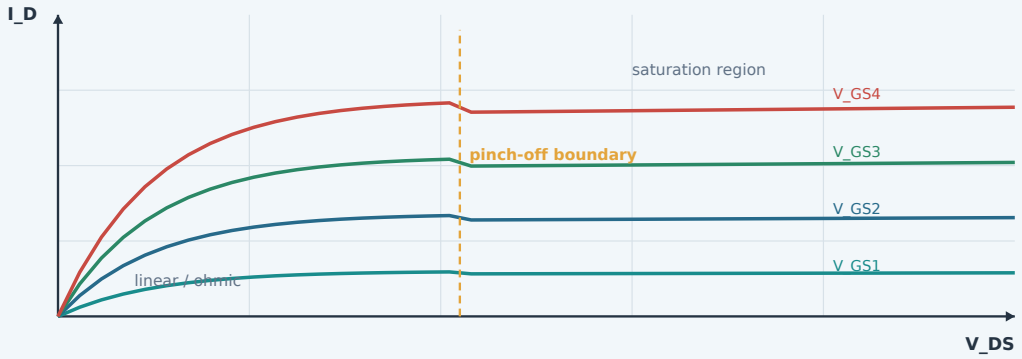
For a long-channel enhancement MOSFET in the ideal square-law model, increasing V_DS gradually reduces channel voltage headroom near the drain. When the drain end reaches threshold, the channel pinches off at the drain end and **$V_{DS,sat} \approx V_{GS} - V_T$** (for $V_{GS} > V_T$). In the saturation region, I_D is approximately set by V_GS and is relatively insensitive to further V_DS increase in the ideal model:

$I_{D,sat} \approx \frac{1}{2} k_n (V_{GS} - V_T)^2$, where $k_n = \mu_n C_{ox}(W/L)$ under one common convention. Channel-length modulation makes real current rise somewhat with V_DS. **Pinch-off does not mean I_D becomes zero**; carriers continue to be swept through the high-field pinch-off region.

For an n-channel JFET, pinch-off convention is often described using a negative gate pinch-off voltage V_P. The saturation boundary is approximately $V_{DS(sat)} \approx V_{GS} - V_P$ with that convention, and Shockley’s transfer relation in its usual magnitude form is $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ over the stated operating range. Different texts define the sign of V_P differently, so copy the convention given. Do not apply the MOSFET threshold formula blindly to a JFET.

“**Saturation voltage**” here means the drain-source voltage at the boundary where the output characteristic enters its saturation/current-source region. It is **not** the transistor’s destructive breakdown voltage. In BJT terminology, “saturation” means both junctions forward biased and is a different operating region.

FET output characteristics and pinch-off



3.3 Reading the FET output characteristic

Plot I_D vertically against V_DS horizontally for several fixed V_GS values. At small V_DS, the channel behaves roughly resistively (ohmic/linear region). After the pinch-off boundary V_DS,sat, curves flatten into the saturation region. Increasing V_GS above threshold raises the drain current. Mark cutoff, linear/ohmic region, pinch-off boundary, saturation region, axes, and the family parameter V_GS. Label the separate high-field breakdown region only as a limit, not as ordinary saturation.

4. Exam graph-drawing checklist

Use this short checklist before you put down the pencil:

Graph / diagram	Axes or labels that must be present	Shape or key labels to include
Diode V–I	I_D vertical, V_D horizontal; forward and reverse signs	forward knee, leakage, breakdown voltage and rise
Diode load line	I_D, V_D; intercept values	straight negative-slope line; intersection/Q point
HWR/FWR waveforms	time t, input and output	zero/negative half-cycle for HWR; both positive lobes for FWR
Capacitor-filter output	time t and output voltage	high DC level, ripple, capacitor charge peaks
BJT output characteristic	I_C, V_CE; I_B values	cutoff, saturation knee, active family, breakdown limit
CE input characteristic	I_B, V_BE; V_CE fixed	diode-like rising curves
CB / CC characteristics	state current/voltage and held-constant variable	use the sign/axis convention in the question
BJT DC + AC load lines	I_C, V_CE; Q point	label both lines; AC line passes through Q
CE frequency response	gain (or gain dB) vs log f	low-frequency rise, midband, f_L at −3 dB
FET output family	I_D, V_DS; V_GS values	linear region, pinch-off/saturation, cutoff and limits

Diagram marking habit: first draw the axes or circuit outline, then label every terminal and polarity, then draw the curves/arrows, then add the operating region and a one-line explanation. A beautiful unlabeled graph is still an incomplete answer.

5. Rapid formula sheet and memory cues

5.1 Diodes and rectifiers

Topic	Formula / reminder
Thermal voltage	$V_T = kT/q \approx 25.85$ mV at 300 K
Diode equation	$I_D = I_S[\exp(V_D/(nV_T)) - 1]$
Inverse diode equation	$V_D = nV_T \ln(1 + I_D/I_S)$
Static resistance	$R_D = V_D/I_D$
Small-signal resistance	$r_d \approx nV_T/I_D$ (forward bias)
Junction capacitance	$C_T \approx C_{J0}/(1 + V_R/\Phi_0)^m$
Diffusion capacitance	$C_D \approx \tau_T I_D/(nV_T)$
Series diode/load line	$I_D = (V_S - V_D)/R$
HWR average / RMS	$V_{DC} = V_m/\pi$; $V_{RMS} = V_m/2$
FWR average / RMS	$V_{DC} = 2V_m/\pi$; $V_{RMS} = V_m/\sqrt{2}$
Ripple frequency	HWR: f; FWR: 2f
Capacitor ripple estimate	$\Delta V_{pp} \approx I_L/(f_r C)$
Zener current	$I_Z = (V_S - V_Z)/R_S - I_L$
Zener power	$P_Z = V_Z I_Z$; always check rating and worst case

Memory cue: HWR has **one** conducting half-cycle; FWR has **two**. Full-wave average is **2/π**, half-wave is **1/π**. Filtering raises the DC level toward the peak and lowers ripple.

5.2 BJT and amplifiers

Topic	Formula / reminder
Current sum	$I_E = I_C + I_B$
Current gains	$\alpha = I_C/I_E$; $\beta = I_C/I_B$; $\gamma = I_E/I_B = \beta + 1$
Convert gains	$\beta = \alpha/(1 - \alpha)$; $\alpha = \beta/(\beta + 1)$
Active-mode current	$I_C \approx \beta I_B$ (not in saturation)
Fixed bias	$I_B \approx (V_{CC} - V_{BE})/R_B$; $V_{CE} = V_{CC} - I_C R_C$
Emitter-resistor bias	$I_B \approx (V_{TH} - V_{BE})/[R_{TH} + (\beta + 1)R_E]$
Divider Thevenin source	$V_{TH} = V_{CC} R_2/(R_1 + R_2)$; $R_{TH} = R_1 \parallel R_2$
CE DC load line	$V_{CE} \approx V_{CC} - I_C(R_C + R_E)$, if $I_E \approx I_C$
AC load	$R_{ac} \approx R_C \parallel R_L$; AC line passes through Q
CE midband gain	$A_v \approx -g_m(R_C \parallel R_L)$; $g_m \approx I_C/V_T$
Capacitor reactance	$X_C = 1/(2\pi fC)$
RC corner	$f_c \approx 1/(2\pi R_{eq} C)$
Voltage gain in dB	$20 \log_{10}(\text{abs}(A_v))$

Memory cue for configurations: CB—α, low input; CE—β, phase flips; CC—β+1, voltage follows. The common terminal is B, E, or C respectively.

5.3 FET syllabus tail

- Enhancement MOSFET channel appears when V_{GS} exceeds V_T.
- Ideal long-channel boundary: $V_{DS,sat} \approx V_{GS} - V_T$.
- Ideal saturation current: $I_{D,sat} \approx \frac{1}{2}k_n(V_{GS} - V_T)^2$.
- JFET already has a channel at zero gate bias; reverse gate voltage narrows it. Do not mix its V_P convention with MOSFET V_T.

6. Practice questions with answers

Try each question before reading its answer. For a 2-mark answer, aim for a precise definition plus one equation/feature. For a longer answer, include a labelled graph or circuit where requested.

6.1 Quick recall — Module 1

M1-Q1. What is the depletion region? Answer: The junction region depleted of mobile carriers after diffusion and recombination leave fixed ionized dopants; its electric field creates the potential barrier.

M1-Q2. What changes in forward bias? Answer: The applied voltage reduces the barrier and depletion width, allowing strong majority-carrier injection and forward current.

M1-Q3. State the diode equation and thermal voltage at room temperature. Answer: $I_D = I_S[\exp(V_D/(nV_T))-1]$; $V_T \approx 25.85$ mV at 300 K.

M1-Q4. Distinguish static and dynamic diode resistance. Answer: Static $R_D = V_D/I_D$ at a point; dynamic $r_d = dV_D/dI_D$ locally, approximately nV_T/I_D in forward bias.

M1-Q5. Which diode capacitance is important under reverse bias? Answer: Depletion/transition capacitance; it generally decreases as reverse bias widens the depletion region.

M1-Q6. Define a load line. Answer: The straight-line voltage-current constraint imposed by the external source and load; its intersection with the device characteristic gives the operating point.

M1-Q7. State the key physical difference between Zener and avalanche breakdown. Answer: Zener-dominant breakdown is tunnelling in a heavily doped, narrow junction; avalanche is impact ionization in a wider, usually more lightly doped region.

M1-Q8. Why is a series resistor essential in a Zener regulator? Answer: It limits current and absorbs excess source voltage so the Zener stays within its allowable current and power range.

M1-Q9. Compare HWR and FWR ripple frequency. Answer: HWR ripple frequency is f; FWR is 2f.

M1-Q10. What does a capacitor filter do? Answer: It charges near waveform peaks and discharges through the load between peaks, raising the average output and reducing ripple.

6.2 Quick recall — Module 2

M2-Q1. Why is the BJT base thin and lightly doped? Answer: To reduce recombination so most carriers injected by the emitter cross the base and reach the collector; only a small base current is required.

M2-Q2. State the BJT current relation. Answer: $I_E = I_C + I_B$.

M2-Q3. Relate α and β. Answer: $\beta = \alpha/(1-\alpha)$, and $\alpha = \beta/(\beta+1)$.

M2-Q4. What are the junction biases in forward-active NPN operation? Answer: Base–emitter forward biased and collector–base reverse biased.

M2-Q5. Which configuration has voltage phase inversion? Answer: CE; its output voltage is approximately 180° out of phase with the input.

M2-Q6. Why is CC called an emitter follower? Answer: The emitter output follows the base input in voltage (with a small V_{BE} offset), giving voltage gain near one and useful current gain.

M2-Q7. What does the Q point represent? Answer: The transistor's DC operating point before the AC signal is applied, commonly specified by I_{CQ} and V_{CEQ}.

M2-Q8. Why does an emitter resistor improve bias stability? Answer: Increased current raises emitter voltage, reducing V_{BE} and opposing the original current increase—negative feedback.

M2-Q9. Why can AC and DC load lines differ? Answer: Coupling capacitors block DC but pass midband AC, so an external load may affect the AC load but not the DC path.

M2-Q10. Name the three capacitors commonly responsible for low-frequency CE roll-off. Answer: Input coupling, output coupling, and emitter-bypass capacitors.

6.3 “Give reasons” — high-yield explanations

1. **A silicon diode is often treated as a 0.7 V drop in hand calculations.** Because its forward characteristic is steep in ordinary operating ranges and a constant-drop model simplifies circuit analysis; the real drop varies with current, temperature, and device.
2. **A diode’s reverse current increases sharply at breakdown.** Strong electric fields cause tunnelling and/or impact ionization, producing many carriers; current must be externally limited.
3. **Zener breakdown often has a negative temperature coefficient while avalanche has a positive one.** Their dominant physical mechanisms respond oppositely to temperature; real coefficients depend on voltage and device, and both mechanisms can contribute.
4. **A full-wave rectifier gives a larger average output than a half-wave rectifier for the same peak input.** It uses both halves of the waveform, so the magnitude is nonzero twice each cycle; the ideal average is $2V_m/\pi$ rather than V_m/π.
5. **A capacitor filter reduces ripple more effectively at higher frequency or larger capacitance.** It has less time to discharge between peaks, and $\Delta V_{pp} \approx I_L/(f_r C)$.
6. **The BJT collector current can be much larger than the base current.** The thin, lightly doped base allows most emitter-injected carriers to be swept into the collector; the base current replenishes only the small recombination fraction.
7. **CE output is inverted.** A rise in base drive raises collector current; the larger drop across R_C lowers collector voltage.
8. **A fixed-bias transistor can drift as temperature or β changes.** Its base current is set chiefly by a resistor and voltage, with little negative feedback to counter changes in collector current.
9. **An emitter resistor stabilizes collector current.** A current rise raises V_E, reduces V_{BE}, and counteracts the rise.
10. **A coupling capacitor affects low-frequency gain more than midband gain.** Its reactance grows as frequency falls, causing larger signal loss at low frequency.
11. **CC has voltage gain close to one but can still be useful.** Its high input impedance, low output impedance, and high current gain make it a buffer/impedance matcher.
12. **Pinch-off in an ideal MOSFET does not stop drain current.** The channel pinches off at the drain end, but carriers are swept across the high-field region; current becomes comparatively insensitive to further V_{DS}.

6.4 Worked numericals — additional practice

N1. Diode equation with a changed voltage

At 300 K, $I_S = 1$ pA, $n = 1$, $V_D = 0.55$ V. Estimate I_D. **Solution:** $x = 0.55/0.02585 \approx 21.28$; $e^x \approx 1.74 \times 10^9$. $I_D \approx 1\text{ pA} \times 1.74 \times 10^9 =$ **1.74 mA** under the ideal equation. Real series resistance and device details can shift the measured value, but the calculation is now in a plausible introductory-circuit range.

N2. Rectifier output and PIV

A 50 Hz, 10 V RMS source feeds an ideal bridge. Find peak input, unfiltered average output, ripple frequency, and per-diode PIV. **Solution:** $V_m = \sqrt{2}(10) =$ **14.14 V**; $V_{DC} = 2V_m/\pi =$ **9.00 V**; ripple frequency = **100 Hz**; bridge PIV per diode ≈ **14.14 V** (ideal, with V_m defined across the complete bridge input).

N3. Required filter capacitance

A 100 Hz full-wave ripple supply must have $\Delta V_{pp} \le 0.5$ V at $I_L = 50$ mA. Find the approximate minimum C. **Solution:** $C \ge I_L/(f_r \Delta V) = 0.050/(100 \times 0.5) = 0.001$ F = **1000 µF**. Choose the next appropriate standard value and verify ripple/current ratings in a real design.

N4. Transistor gain conversion

A transistor has $\alpha = 0.98$. Find β. **Solution:** $\beta = \alpha/(1-\alpha) = 0.98/0.02 =$ **49**.

N5. Fixed-bias active-region check

For $V_{CC} = 9$ V, $R_C = 1$ kΩ, $I_C = 4$ mA, estimate V_{CE} with grounded emitter. **Solution:** $V_{CE} = 9 - (0.004)(1000) =$ **5 V**. This is not near the saturation knee, so the assumed active condition is plausible, subject to the actual circuit.

N6. CE low-frequency corner

A coupling capacitor C = 10 μF sees an effective resistance of 2 kΩ. Estimate its corner. **Solution:** $f_c = 1/[2\pi(2000)(10 \times 10^{-6})] \approx$ **7.96 Hz**. This is one pole's estimate; the amplifier's overall f_L depends on all low-frequency capacitors.

6.5 Longer exam-style prompts and answer plans

L1. Explain a PN junction under zero, forward, and reverse bias; draw and explain its V–I characteristic. Answer plan: define P and N doping; explain diffusion, recombination, depletion region and built-in field; zero-bias equilibrium; forward bias narrows barrier and causes carrier injection; reverse bias widens it and leaves small leakage; draw I_D against V_D with forward knee, reverse leakage, breakdown and clear polarity labels. Mention the approximate model only with a caveat.

L2. Explain the operation of a Zener shunt regulator and design its series resistor. Answer plan: draw source–R_S–output node, reverse-connected Zener and load in parallel; explain current division $I_R = I_L + I_Z$; derive $R_S = (V_S - V_Z)/(I_L + I_Z)$ at a chosen minimum I_Z; check minimum source/maximum load for regulation and maximum source/minimum load for Zener/resistor power.

L3. Compare half-wave and full-wave rectifiers, with circuits, waveforms, equations, and a capacitor filter. Answer plan: show the HWR diode/load and one conducting half-cycle; show either CT or bridge FWR and both positive lobes; give V_{DC}, V_{RMS}, ripple frequency, PIV and conducting-path drops with V_m defined; show capacitor charge/discharge and ΔV_{pp} estimate; conclude why FWR is easier to filter.

L4. Explain basic transistor action and compare CB, CE, and CC. Answer plan: draw NPN/PNP symbols and arrow rule; explain active junction biases and thin base; give $I_E = I_C + I_B$ and α/β relations; table the common terminal, input/output ports, gain, resistance, phase, and use; distinguish qualitative trends from exact values.

L5. Explain fixed bias and emitter/self-bias; find the Q point. Answer plan: draw each circuit; write base-loop and collector-loop equations; state $I_C \approx \beta I_B$ only in active mode; calculate V_{CE} and check saturation; explain negative feedback from R_E and why divider bias is less β-sensitive. If "self-bias" terminology is ambiguous, state which circuit you mean.

L6. Draw and explain CE input/output characteristics and DC/AC load lines. Answer plan: label axes and held-constant variables; show input I_B–V_{BE}; show output curves for fixed I_B with cutoff, saturation knee, active region; draw DC line, mark Q, then AC line through Q with slope set by $R_C||R_L$; explain signal swing and clipping.

L7. Explain the low-frequency response of a CE amplifier. Answer plan: draw the CE stage with input/output coupling and emitter bypass capacitors; explain X_C increases as f decreases; describe each capacitor's effect; sketch gain against log f with low-frequency rise, midband plateau and f_L at −3 dB; give $f_c \approx 1/(2\pi R_{eq}C)$ and state that overall f_L depends on all poles.

L8. Explain FET channel formation, pinch-off, saturation voltage, and output curves. Answer plan: first name enhancement MOSFET or JFET; explain gate-field control; draw I_D–V_{DS} families for several V_{GS}; label linear region and saturation/pinch-off boundary; give $V_{DS,sat} \approx V_{GS} - V_T$ for the ideal enhancement MOSFET; state pinch-off does not make I_D zero; distinguish saturation from breakdown and from BJT saturation.

7. Balanced 100-mark mock paper and marking guide

Practice allocation: Module 1 = 50 marks; Module 2 = 50 marks, matching your 50:50 revision note. This is a self-made practice paper, not a prediction of your actual examination. Suggested practice: attempt closed-book, label every diagram, and then mark using the points below.

Section A — short answers (10 × 2 = 20 marks)

1. Define the diode depletion region. (M1, 2)
2. Write the diode equation and give V_T at 300 K. (M1, 2)
3. State one physical difference between Zener and avalanche breakdown. (M1, 2)
4. Give the ideal ripple frequency of a full-wave rectifier on f-Hz AC. (M1, 2)
5. State one effect of a capacitor filter. (M1, 2)
6. State the BJT current relation and define β. (M2, 2)
7. Name the junction biases for forward-active NPN operation. (M2, 2)
8. Which BJT configuration gives 180° voltage phase inversion? (M2, 2)
9. What is the Q point? (M2, 2)
10. Give one reason an emitter resistor stabilizes bias. (M2, 2)

Section B — structured answers (8 × 5 = 40 marks)

11. Draw and explain the diode V–I characteristic, marking forward knee, reverse leakage, and breakdown. (M1, 5)
12. Compare HWR and bridge FWR by output waveform, average output, ripple frequency, and diode drops/PIV convention. (M1, 5)
13. Explain a Zener shunt regulator and write the current and power checks required. (M1, 5)
14. A 50 Hz, 12 V RMS source feeds an ideal full-wave rectifier. Find V_m, unfiltered V_{DC}, and ripple frequency. (M1, 5)
15. Explain transistor action and relate α, β, I_B, I_C, and I_E. (M2, 5)
16. Compare CB, CE, and CC using common terminal, current gain, input resistance, phase, and application. (M2, 5)
17. For fixed bias, V_{CC} = 10 V, R_B = 465 kΩ, R_C = 2 kΩ, β = 100, V_{BE} = 0.70 V. Find I_B, I_C, and V_{CE}; check active-mode plausibility. (M2, 5)
18. Explain the three main low-frequency causes of gain loss in a CE amplifier and sketch the response. (M2, 5)

Section C — long answers (4 × 10 = 40 marks)

19. Explain HWR and FWR with circuits and output waveforms. Derive or state average and RMS values, ripple frequency, PIV for HWR and bridge FWR, and explain the capacitor filter with its ripple estimate. (M1, 10)
20. A Zener regulator uses V_S = 15 V, V_Z = 6.2 V, R_L = 1.0 kΩ, and R_S = 560 Ω. Assuming ideal source and nominal Zener voltage, calculate load current, series current, Zener current, and Zener power. State why a worst-case rating check is still required. (M1, 10)
21. Draw CE output characteristics, explain cutoff/saturation/active regions, construct DC and AC load lines, and identify the Q point and signal-swing limits. (M2, 10)
22. Explain fixed bias and emitter/divider self-bias; calculate a fixed-bias Q point for V_{CC} = 12 V, R_B = 470 kΩ, R_C = 2.2 kΩ, β = 100, V_{BE} = 0.70 V. Then explain why adding R_E improves stability. (M2, 10)

Marking guide / answer key

A1: region depleted of mobile carriers, with fixed ionized dopants/electric field (2). **A2:** $I_D = I_S[\exp(V_D/(nV_T))-1]$ (1); $V_T \approx 25.85$ mV at 300 K (1). **A3:** tunnelling/heavily doped narrow junction versus impact ionization/wider junction (2). **A4:** 2f (2). **A5:** raises average output/reduces ripple by charging near peaks and discharging into the load (2). **A6:** $I_E = I_C + I_B$ (1); $\beta = I_C/I_B$ (1). **A7:** EB forward, CB reverse (2). **A8:** CE (2). **A9:** DC quiescent operating point, usually I_{CQ} and V_{CEQ} (2). **A10:** rising current raises V_E and reduces V_{BE}, creating negative feedback (2).

B11 (5): correct axes/signs (1); forward knee and steep rise (1); reverse leakage (1); breakdown and label (1); clear curve/explanation (1). **B12 (5):** HWR one lobe, FWR both (1); correct averages V_m/π and $2V_m/\pi$ (1); f and 2f (1); bridge two diode drops (1); bridge PIV V_m when V_m is complete input peak (1). **B13 (5):** correct reverse shunt circuit (1); $V_{OUT} \approx V_Z$ (1); $I_R = I_L + I_Z$ (1); R_S/current design (1); Zener/resistor power checks and ratings (1). **B14 (5):** $V_m = 12\sqrt{2} = 16.97$ V (2); $V_{DC} = 2V_m/\pi = 10.80$ V (2); ripple = 100 Hz (1). **B15 (5):** EB forward/CB reverse (1); thin/light base and carrier injection (1); most carriers collected (1); $I_E = I_C + I_B$ (1); α/β relation or active-mode $I_C \approx \beta I_B$ (1). **B16 (5):** common terminals (1); gains CB α, CE β, CC $\beta+1$ (1); resistance tendencies (1); CE inversion vs CB/CC (1); suitable uses (1). **B17 (5):** $I_B = (10-0.70)/465k = 20$ µA (2); $I_C = 2.0$ mA (1); $V_{CE} = 10-(2\text{mA})(2\text{k}) = 6.0$ V (1); active plausible (1). **B18 (5):** identify input, output, emitter bypass capacitors (3); explain high X_C at low f and reduced gain (1); response sketch with midband/f_L (1).

C19 (10): both circuits and conduction paths (2); correct output shapes (2); averages/RMS (2); ripple frequency and PIV definitions (2); capacitor action and ΔV_{pp} equation (2). **C20 (10):** $I_L = 6.2$ mA (2); $I_R = (15-6.2)/560 = 15.71$ mA (3); $I_Z = 9.51$ mA (2); $P_Z = 6.2 \times 9.51\text{mA} \approx 58.9$ mW (1); explain max/min source/load, tolerances and power/current rating checks (2). **C21 (10):** axes and IB curve family (2); label cutoff, saturation, active region (2); DC load line (2); AC line through Q and $R_C \| R_L$ slope (2); Q and clipping/swing limits (2). **C22 (10):** fixed-bias circuit/equations (2); correct calculations (3); determine active point (1); emitter feedback mechanism (2); divider Thevenin/β-insensitivity or comparison (2).

Mock calculation checks: Q17: $I_B = 9.3/465k = 20$ µA; $I_C = 2.0$ mA; $V_{CE} = 6.0$ V. Q20: $I_L = 6.2$ mA; $I_R = 8.8/560 = 15.714$ mA; $I_Z = 9.514$ mA; $P_Z \approx 0.059$ W. Q22: $I_B = 11.3/470k = 24.0$ µA; $I_C \approx 2.40$ mA; $V_{CE} = 12 - 2.40\text{mA} \times 2.2\text{k} = 6.72$ V.

8. Final self-test and last-night plan

8.1 One-minute closed-book checklist

Can you do these without notes?

- Explain PN-junction formation and forward/reverse bias in three sentences.
- Write the diode equation, calculate V_T, distinguish R_D from r_d, and explain the two diode capacitances.
- Draw diode I–V and a load line with correct axes and Q point.
- Explain Zener versus avalanche and calculate a Zener series resistor/current.
- Draw HWR and bridge FWR, their output waveforms, and capacitor ripple; distinguish V_m, V_RMS, V_DC, f, and 2f.
- Explain NPN transistor action, write all current-gain relations, and name cutoff/active/saturation.
- Compare CB, CE, CC; sketch CE input/output characteristics and explain the phase reversal.
- Calculate a fixed-bias Q point and explain why emitter/divider bias is more stable.
- Draw DC and AC load lines and identify the Q point.
- Sketch a CE low-frequency response with the three capacitors and f_L.
- Explain why channel/pinch-off language is FET terminology and give the MOSFET saturation boundary.

Any “no” is a targeted review item—not a reason to reread the whole guide.

8.2 A practical 90-minute final review

This is a suggestion, not a guarantee or a substitute for sleep:

- **0–10 min:** scan the formula sheet; write key equations from memory.
- **10–30 min:** Module 1: redraw diode curve, load line, Zener regulator, HWR/FWR waveforms; solve one diode and one rectifier numerical.
- **30–50 min:** Module 2: redraw NPN symbol, configuration comparison, CE characteristics, DC/AC load lines; solve one bias problem.
- **50–65 min:** do the “give reasons” questions aloud; repair weak explanations.
- **65–80 min:** attempt selected mock-paper questions without notes; check units and marks.
- **80–90 min:** review only mistakes and the graph checklist, pack what you need, and stop. A rested brain retrieves better than an all-night reread.

8.3 Final exam-answer checklist

Before submitting an answer, check: **Did I define the term? Label the diagram? State the assumed diode/transistor model? Use the correct sign and units? Distinguish DC from AC? Explain the graph’s regions? Check whether the result is physically plausible?** If yes, you have turned knowledge into marks.

You now have one complete coverage-aligned revision path for every topic in the syllabus you pasted, plus the exam-emphasis list and a carefully separated FET clarification. Focus on retrieval and labelled drawings—not on trying to memorize this guide word for word.

Source and scope record

Coverage source: the user-provided syllabus for “(EC 101) Basics of Electronics & Communication Engineering,” Module 1 and Module 2, plus the seven-item “ECE EXAM IMPORTANT TOPICS” list in the request. The guide extends explanations and practice for learning purposes; it does not claim access to additional materials or prediction of a particular exam.